Applied Physics Letters

In-plane magnetization and electronic structures in $BiFeO_3$/graphene superlattice

Cite as: Appl. Phys. Lett. **120**, 084002 (2022); https://doi.org/10.1063/5.0083643
Submitted: 28 December 2021 • Accepted: 07 February 2022 • Published Online: 24 February 2022

Chen Chen, Junjie Zeng, Yafei Ren, et al.

View Online Export Citation CrossMark

QBLOX

1 qubit

Shorten Setup Time
Auto-Calibration
More Qubits

Fully-integrated
Quantum Control Stacks
Ultrastable DC to 18.5 GHz
Synchronized <<1 ns
Ultralow noise

100s qubits

visit our website >

© 2022 Author(s).

Applied Physics Letters ARTICLE scitation.org/journal/apl

In-plane magnetization and electronic structures in $BiFeO_3$/graphene superlattice

Cite as: Appl. Phys. Lett. **120**, 084002 (2022); doi: 10.1063/5.0083643
Submitted: 28 December 2021 · Accepted: 7 February 2022 ·
Published Online: 24 February 2022

View Online Export Citation CrossMark

Chen Chen,[1,2] Junjie Zeng,[3] Yafei Ren,[3,4] Le Fang,[1,2] Yabei Wu,[1,5] Peihong Zhang,[1,5] Tao Hu,[1] Jian Wang,[6,a)] Zhenhua Qiao,[3,4,a)] and Wei Ren[1,2,7,a)]

AFFILIATIONS

[1]International Center for Quantum and Molecular Structures, Materials Genome Institute, Physics Department, Shanghai University, Shanghai 200444, China
[2]Shanghai Key Laboratory of High Temperature Superconductors, Shanghai University, Shanghai 200444, China
[3]ICQD, Hefei National Laboratory for Physical Sciences at Microscale, and Synergetic Innovation Center of Quantum Information and Quantum Physics, University of Science and Technology of China, Hefei, Anhui 230026, China
[4]CAS Key Laboratory of Strongly-Coupled Quantum Matter Physics, and Department of Physics, University of Science and Technology of China, Hefei, Anhui 230026, China
[5]Department of Physics, University at Buffalo, State University of New York, Buffalo, New York 14260, USA
[6]College of Physics and Optoelectronic Engineering, Shenzhen University, Shenzhen, Guangdong, China
[7]Zhejiang Laboratory, Hangzhou 311100, China

[a)]**Authors to whom correspondence should be addressed:** jianwang@hku.hk; qiao@ustc.edu.cn; and renwei@shu.edu.cn

ABSTRACT

We predict the presence of gapped electronic structures in an artificial superlattice of graphene embedded in (111)-oriented $BiFeO_3$ layers based on first-principles calculations. Due to the electron transfer and the proximity effect at the $BiFeO_3$/graphene interface, we find that magnetic moments of Fe atoms near a graphene layer were slightly less than that of bulk Fe atoms. Regarding the ferromagnetic moment orientation of Fe atoms in perovskite $BiFeO_3$, we reveal that the in-plane magnetization gives the ground state. The bandgap depends on the magnetization direction and the separation between the graphene layer and the perovskite $BiFeO_3$ slab, which might be adjusted by applying external uniaxial stress in an experiment. Our results provide a route for designing hybrid 2D materials with emerging properties that are not available in single materials alone.

Published under an exclusive license by AIP Publishing. https://doi.org/10.1063/5.0083643

In 1879, the Hall effect was discovered when a perpendicular magnetic field was applied to an electric current in a metal sheet, in which the Lorentz force was generated onto electrons.[1] In clean samples of a low-density two-dimensional electron gas, discrete Landau levels are created by a strong magnetic field leading to the quantized Hall conductance, known as the "quantum Hall effect" (QHE).[2] Such intriguing effects are important not only for the fundamental breakthrough to bring topology to the condensed matter physics but also for its potential practical application in dissipationless electronic devices. However, the prerequisite of the strong magnetic field limits the application of QHEs. In 1988, Haldane suggested the possibility to realize QHEs in the absence of a magnetic field by introducing a staggered magnetic flux in a hexagonal honeycomb lattice system. To distinguish it from the traditional QHE from the strong external magnetic field, this phenomenon is named as the "quantum anomalous Hall effect" (QAHE).[3,4]

Nevertheless, the staggered magnetic flux is also difficult to be achieved experimentally that hinders the observation of QAHEs. Over recent decades, great progress has been made in exploring how to achieve QAHEs in realistic two-dimensional materials both theoretically and experimentally. The proposals mainly focus on two categories, i.e., magnetically doped topological insulator thin films and magnetized narrow-gapped two-dimensional atomic-layer crystals. In experiments, the former strategy has been realized experimentally in a magnetically doped topological insulator film Cr-doped $(Bi, Sb)_2Te_3$ sample by using molecular beam epitaxy growth.[5] However, the observations are made under extremely low temperatures limited by the low Curie temperature and small energy gap. To take advantage of QAHEs in the dissipationless electronic devices, the presence of QAHEs at higher temperatures is strongly demanded, which is still a difficult challenge awaiting to be solved.

Published under an exclusive license by AIP Publishing

Graphene provides another possible solution as a unique two-dimensional material.[6] Graphene has attracted tremendous attention because of its superior mechanic, thermal, optical, and electrical properties,[7,8] including the linear Dirac dispersion relationship[6] and extremely high carrier mobility.[6,8] Although the pristine graphene is gapless and does not possess either magnetization or strong spin–orbit coupling, the QAHE might be induced by the externally tunable magnetization and Rashba spin–orbit coupling. To introduce magnetization and Rashba spin–orbit coupling, one may try doping magnetic metal atoms on graphene[9,10] or by placing the graphene on magnetic substrates.[11–13] Although the atomic adsorption is a simple and straightforward scheme, the metal atoms are likely to form clusters on the surface of graphene,[14–16] which is detrimental to graphene-based electronic transport. On the other hand, it has been reported that the magnetism in the topological insulators can be induced by the proximity coupling with magnetic insulators.[17–22] For example, Xiao *et al.* predicted that the bilayers of perovskite-type transition-metal oxides grown along the [111] crystallographic axis are potential candidates for two-dimensional topological insulators;[23] Weng *et al.* found the gap opening of Dirac cones by the spin–orbit coupling with a nontrivial topological magnetic phase corresponding to the quantum anomalous Hall effect from the $LaMnO_3$ (111) bilayer sandwiched between $LaScO_3$ barriers;[24] they further proposed strong magnetoelectric couplings for the electric field to control the magnetization in the $BiFeO_3$ (BFO) bilayer sandwiched in $SrTiO_3$ along the [111] axis.[25] Therefore, the antiferromagnetic perovskite oxide $BiFeO_3$ is an ideal choice of a substrate as such an insulating substrate does not affect the electronic transport properties of graphene. Moreover, the Fe atoms on the (111) surface are intra-layer ferromagnetic, but inter-layer antiferromagnetic, resulting in a local magnetic moment but a vanishing overall magnetization. The orbital hybridization of graphene and substrates can also induce a substantial spin–orbit coupling effect, which benefits the presence of a QAHE bandgap. However, the energy gap is located far away from the Fermi level as reported in the previous work in the heterostructure interface of graphene and multiferroic BFO.[11,26–29]

Here, we constructed the superlattices of graphene and BFO[26–29] in this work. Such a hybrid system can be assembled by using frontier fabrication technology of layer-by-layer construction in a precisely controlled sequence.[30,31] The multiferroic perovskite BFO[32] with a bandgap of about 2.7 eV[33] is both ferroelectric (Curie temperature $T_C \cong 1103$ K) and antiferromagnetic. (The Néel temperature T_N is approximately 643 K.)[34] At room temperature, the space group of BFO is R3c with the lattice constants $a = 5.58$ Å and $c = 13.87$ Å.[35] The structure can be described as rhombohedral deformation of cubic perovskite cells with both ferroelectricity and oxygen octahedral tilting along the (111) axis. BFO is a G-type antiferromagnet at room temperature, in which Fe^{3+} ions have large magnetic moment. Along the (111) direction, the Fe ions of the same atomic layer have the same direction of magnetic moment, and the adjacent layers of Fe ions have opposite directions of magnetic moment. Therefore, we construct a (111) terminated BFO/graphene superlattice, which may induce both the exchange field and Rashba spin–orbit coupling. People can prepare heterojunctions of graphene and BFO with the help of state-of-the-art techniques. Recently, Katiyar *et al.* successfully deposited pure phase BFO films on $Pt/TiO_2/SiO_2/Si$ substrates by using pulse laser deposition,[36] and Wu *et al.* stacked BFO nanoplates on monolayer graphene over a Si substrate with a 285 nm SiO_2 layer.[37]

We found that in the relaxed BFO/graphene superlattice, the average distance between graphene and the Fe atomic layer is about 2.94 Å, and the spacing between graphene and the Bi layer is approximately 3.75 Å. The Fe atoms near the graphene layer have magnetic moment (3.97 μ_B) that is slightly smaller than the other Fe atoms (4.16 μ_B). Later, we illustrate the charge density difference of the BFO/graphene superlattice to check the magnetic change and electron transfer. We have also studied the electronic band structures with and without spin–orbit coupling (SOC) for the BFO/graphene superlattice. Through our previous work, it has been proved that there is a quantized anomalous Hall conductance in BFO/graphene heterostructures with out-of-plane magnetization.[11] Therefore, in this work, we pay more attention to the influence of the in-plane magnetization in the following. We revealed the effect of the magnetization of Fe atoms in perovskite BFO on the electronic states located at the interface of the BFO/graphene superlattice system. The SOC band gaps δ of about 1.65, 1.11, and 1.30 meV are found at the K and K' valleys, when the magnetization directions are, respectively, along the X-axis, Y-axis, and Z-axis. By calculating the total energies with different magnetization directions, we confirmed that the system is in the ground state when the direction of the magnetic moment is in-plane, as shown in Table S1 and Fig. S1 of the supplementary material. Finally, we demonstrated how to enhance the Rashba SOC and band gap in graphene by reducing the distance between graphene and BFO.

Our density functional theory (DFT)[38] calculations employed the Perdew–Burke–Ernzerhof (PBE) method as implemented in the Vienna *Ab initio* Simulation Package (VASP).[39] The electron exchange–correlation interaction was described by the generalized gradient approximation (GGA) plus the on-site Coulomb interaction (GGA + U) approach.[40] The effective value of Hubbard U was chosen to be 4 eV on the Fe atoms, and we also considered the van der Waals interaction by using the D3 and optB86b vdW functional in the DFT calculation.[41] The kinetic energy cutoff was chosen as 550 eV for the plane wave basis set, and the first Brillouin-zone integration was $3 \times 3 \times 2$ Monkhorst-Pack grids. The convergence of electronic energy was set to 10^{-6} eV in the self-consistent calculations. During structural optimization, all atoms in the BFO/graphene superlattice were allowed to relax along any direction, and the interatomic forces were converged to less than 0.01 eV/Å.

In our calculations, periodic boundary conditions are applied when we construct the BFO/graphene superlattice by inserting a 4×4 supercell of graphene into the $\sqrt{3} \times \sqrt{3}$ supercell of six-layer (111)-orientated BFO, as shown in Fig. 1(a). The lattice parameters of graphene and BFO allow that the lattice mismatch between graphene and BFO in our calculation is only about 1.78%. The relaxed distance between graphene and the Fe atoms is found to be $d_0 = 2.94$ Å, and that between the graphene and the Bi atomic layer is approximately 3.75 Å as shown in Fig. 1(b). There are three Fe atoms located beneath graphene per unit cell: one below the honeycomb hole (H-site Fe), and the other two are below A-site and B-site Fe, as illustrated in Fig. 1(c).

To show the variation of the charge density before and after insertion of graphene, we plot the charge-density difference of the BFO/graphene superlattice in Fig. 2(a). We can find that the electrons tend to transfer toward the Fe atoms near the graphene layer. Moreover, the electron transfers around the three types of Fe atoms are found to be different in detail. As shown in Figs. 2(b) and 2(c), the top and side views of the charge-density difference illustrate the

Published under an exclusive license by AIP Publishing

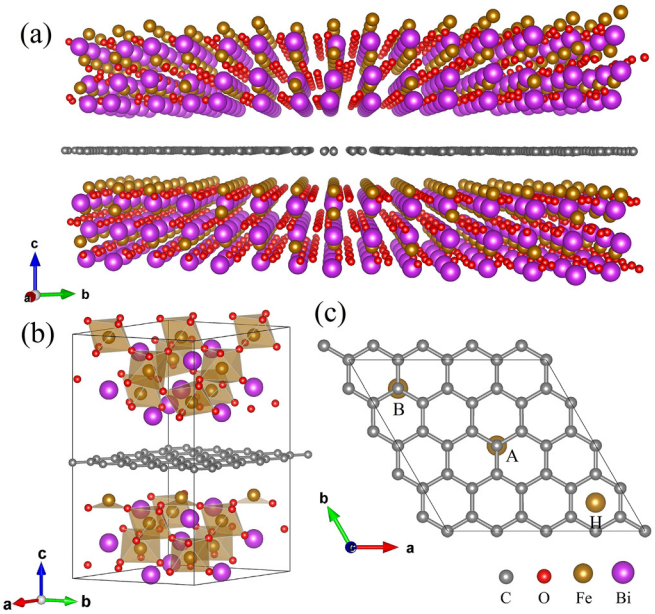

FIG. 1. (a) Perspective view of the BFO/graphene superlattice. Spheres in gray, red, brown, and purple represent carbon (C), oxygen (O), iron (Fe), and bismuth (Bi) atoms, respectively. (b) Optimized structure of the superlattice with graphene embedded into perovskite BFO slabs with the (111) interface. (c) Top view of graphene and three underlying Fe atoms in adjacent layers, where A, B, and H-site Fe atoms are located below two carbon atoms and the honeycomb hole.

visualization of electron accumulation and depletion (yellow and cyan, respectively) among A, B, and H-site Fe atoms and graphene. More importantly, the proximity effect from the magnetized Fe layer is expected to efficiently magnetize the graphene layer.[11]

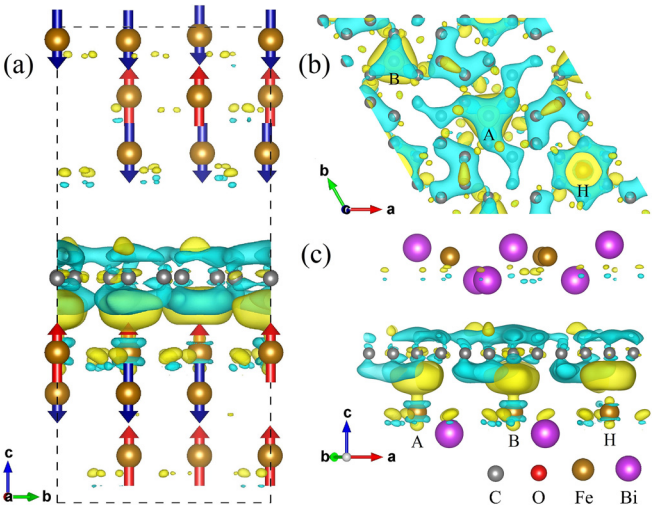

FIG. 2. (a) The charge density difference of the BFO/graphene superlattice, where the yellow and cyan colors indicate electron accumulation and depletion, respectively. Red and blue arrows represent the directions of the magnetic moments of Fe. (b) Top view and (c) side view of the charge density difference show the difference of charge transfer for Fe atoms at A-site, B-site, and H-site.

Then we study the electronic structures of BFO/graphene superlattice by calculating the energy bands along the path Γ-K'-K-Γ of the Brillouin zone, as illustrated in Fig. 3. Figure 3(a) shows the spin-resolved band structure of the superlattice without including the SOC effect. We find that the band structure is basically formed by the superimposition of graphene and perovskite BFO. However, the spin-majority and spin-minority bands of graphene are split as indicated by red and blue, respectively. Figure 3(b) shows the zoom-in of the spin-polarized band structure around the Dirac point near K with an exchange splitting M ∼ 33.11 meV. Because of the magnetic Fe atoms below the graphene layer, the spin-split π bands of graphene have the majority (minority)-spin energy band shifting downward (upward). It is noteworthy that the crossings of the majority-spin and minority-spin bands are located exactly at the Fermi level, which makes the system an insulator when the SOC is included. We also note that due to the perturbation of staggered A and B sublattice potentials in the graphene, at the K point, the gaps (11.32 and 15.48 meV) are seen for the same majority or minority spin bands.

When the spin–orbit coupling is further included, the global band gaps δ are opened at the intersections of spin-majority and spin-minority bands. The global gap values are 1.65, 1.11, and 1.30 meV for the magnetization directions of perovskite BFO aligned in the X-axis, Y-axis, and Z-axis, respectively, as illustrated in Figs. 3 and 4. Figure 3(d) shows the zoom-in of the energy dispersion near the K point of Fig. 3(c), clearly indicating that the Fermi level is inside the

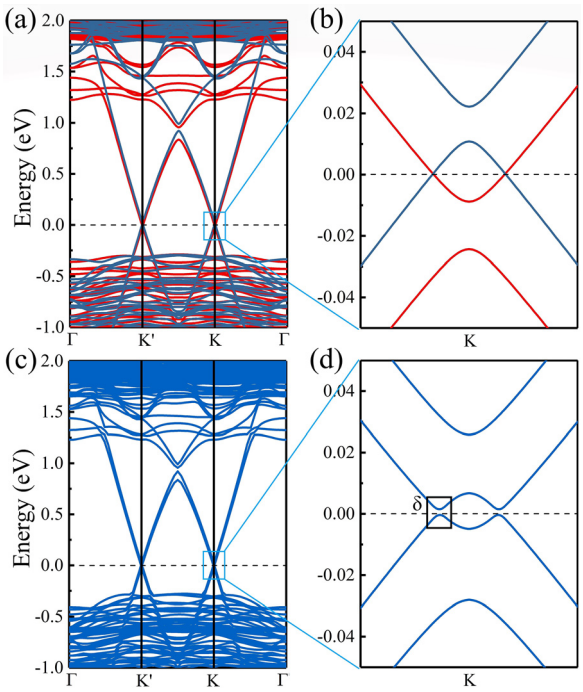

FIG. 3. The electronic band structures of the BFO/graphene superlattice. (a) and (b) Spin-resolved electronic band structures where the red and blue lines are for the majority-spin and minority-spin electrons. (c) and (d) Band structures when SOC is included for the magnetization along the X-axis. The zoom-in energy dispersion near the K point is shown in (b) and (d). The dashed horizontal line at 0 eV indicates the Fermi level in all panels.

Published under an exclusive license by AIP Publishing

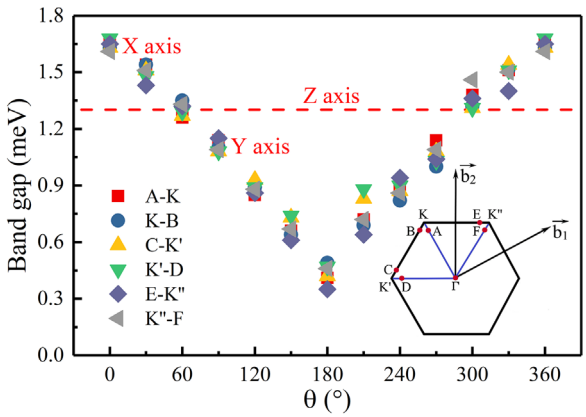

FIG. 4. The relationship between the SOC bandgap and the in-plane magnetizations' rotation angle θ of Fe atoms. The six kinds of symbols indicate the data calculated along the different K-path in the Brillouin zones as shown in the inset. The red dashed line represents the QAHE bandgap of the BFO/graphene superlattice when the magnetization direction is along the Z-axis.

SOC-induced gap and all the band gaps form a ring area around the K point. We have further explored that for the BFO/graphene superlattice, the magnetic moment orientation of Fe atoms has a great influence on the state at the interface in the system. When the magnetic moment is out-of-plane, it gives a Chern insulator; when the magnetic moment is in-plane, the insulator has a Chern number of zero. The magnetic moments tend to be aligned in-plane, and when the magnetization direction is the X-axis, the system has the largest global bandgap of 1.65 meV.[42] In our present superlattice, the magnetic order leads to broken time-reversal symmetry, and the insertion of graphene also destroys mirror symmetries.

The low-energy electronic structure of different magnetization direction can be understood using the following Hamiltonian model:[11]

$$H(\boldsymbol{k}) = -v_f(\eta\sigma_x k_x + \sigma_y k_y)\mathbf{1}_s + \mathbf{1}_\sigma(\mathbf{M}\cdot\mathbf{s}) + \frac{\lambda_R}{2}(\eta\sigma_x s_y - \sigma_y s_x) + U\sigma_z\mathbf{1}_s, \quad (1)$$

where $v_f = 3t/2$ is the Fermi velocity with hopping energy $t \sim 2.60$ eV and $\eta = \pm 1$ are for K and K′ points, respectively. Pauli matrices $\boldsymbol{\sigma}$ and $\mathbf{s}$ stand for sublattices and spin degrees of freedom, respectively. The first term in Eq. (1) describes the Hamiltonian of the pristine graphene, the second term gives the π-electron exchange coupling to magnetic Fe ions, and the third term corresponds to the Rashba SOC.[43] The last term represents the staggered sublattice potential. Both terms of exchange splitting and Rashba SOC play key roles[44–46] in the formation of energy gap at the interface of BFO and graphene. The former term arises from the proximity exchange coupling by magnetic Fe atoms, and the latter term originates from breaking the mirror symmetry of the single layer graphene.[11] The staggered sublattice potential mentioned before is also important in the formation of the bandgap δ when the magnetization lies in plane. We can extract the values of the model parameters by fitting Eq. (1) to the first principles band structure near the Dirac points in Figs. 3(b) and 3(d). For example, the magnetic exchange field in the second term in Eq. (1) for the present BFO/graphene superlattice is estimated to be $M \sim 32.76$, 32.96, and 33.6 meV, when the magnetic moments of perovskite BFO lie in the X-axis, Y-axis and Z-axis, respectively. From the third term, we obtain that the Rashba SOC strength is $\lambda_R \sim 1.94$, 1.31, and 1.53 meV.

The above calculations show that the magnetization orientation of the BFO/graphene superlattice tends to be in-plane. In order to explore its effect on the gap of the superlattice system, we take the X-axis as the initial direction in the XY plane, rotate the magnetization direction of the Fe atoms counterclockwise, and use the parameter θ to indicate the angle between the magnetization and the initial direction, i.e., when θ is 0°, the magnetization orientation is the X-axis, and 90° is for the Y-axis. We can see that the band gap depends on the in-plane magnetization angle θ, being symmetric with respect to the X-axis, as illustrated in Fig. 4. From the X-axis to the −X-axis of the system, the bandgap gradually decreases, reaching a minimum of 0.43 meV for the −X-axis. This is due to the broken spatial inversion symmetry owing to insertion of the graphene layer in the perovskite BFO. Using this special result, one can control the gap quantitatively at the graphene layer by adjusting the in-plane magnetization orientation of the Fe atoms.

To further enlarge the energy gap, we try to enhance the proximity effect between graphene and the BFO slab by simply reducing the spacing between graphene and the perovskite BFO since both the exchange field and the Rashba SOC are caused by hybridization between them. In the experiment, one can control the distance between the graphene and the BFO slab by applying an external uniform stress perpendicular to the graphene layer. Here, we used a way to simulate the effect of stress on the BFO-graphene superstructure; in the case, we moved only the graphene layer to the adjacent Fe atomic layer, while keeping all other atomic layers fixed.

We plot in Fig. 5 for the magnetic moment of three different types of Fe atoms (A, B, and H-sites) and total magnetic moment for the whole BFO/graphene superlattice as a function of the distance change (d_0–d). When the separation between the graphene layer and the Fe atomic layer is reduced by 0.8 Å, we find that the A, B, and H-site Fe magnetic moments decrease by 0.081 μ_B, 0.050 μ_B, and 0.019 μ_B,

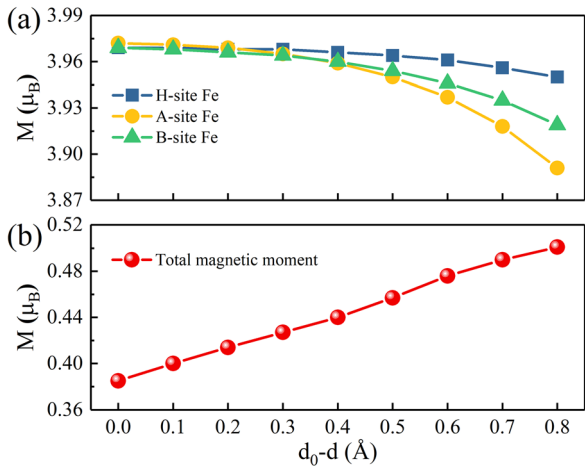

FIG. 5. The calculated (a) local magnetic moments of Fe and (b) total magnetic moments as a function of the displacement (d_0–d) in the BFO/graphene superlattice (d_0 represents the relaxed distance between the graphene layer and the underlying Fe atoms obtained after the geometry optimization).

Published under an exclusive license by AIP Publishing

respectively. The total magnetic moment increases by 0.116 μ_B for the whole system. As the graphene monolayer gets closer to the Fe atomic layer, enhanced electron charge transport from graphene to the Fe layer results in the slight reduction of Fe atomic magnetic moments. Consistent with our analysis of the charge density difference, the hybridization between graphene and A-site Fe is most affected, and the H-site Fe is least influenced by the distance change.

We further explore the dependence of energy gap δ, the Rashba SOC strength λ_R, and the exchange field M on the BFO-graphene distance (d_0–d) as shown in Fig. 6. We find that when the interlayer distance is reduced, the values of δ, λ_R, and M are all increased monotonically. It is noteworthy that, when we reduce the Fe–graphene layer distance by 0.8 Å with magnetization along the X-axis, i.e., interlayer distance d = 2.14 Å, the bandgap δ, the Rashba SOC strength λ_R, and the exchange field M can reach 2.89, 3.40, and 125.63 meV, respectively. When we set the Fe atomic magnetic moments in the Y-axis and reduce the Fe-graphene layer separation by 0.8 Å, the bandgap δ, the Rashba spin–orbit coupling constant λ_R, and the exchange field M increase to 2.15, 2.53, and 125.53 meV, respectively. From comparison of these model magnetizations, we find obvious enhancement effects at the graphene layer inside the BFO/graphene system regardless of the Fe atoms' magnetization direction.

Our simulations show that the stress can effectively enhance the magnetization strength, Rashba SOC, and thus, the energy gap. The most stress sensitive parameter is the magnetization strength that changes dramatically as a result of an interlayer distance reduction of 0.8 Å. As a final remark, we further considered the van der Waals interaction, and the distance between graphene and the Fe atomic layer (Bi atomic layer) can be slightly decreased from d_0 = 2.94 Å (3.75 Å) to d_0 = 2.60 Å (3.28 Å), which simply helps us obtain the larger bandgap δ and the Rashba spin–orbit coupling λ_R.

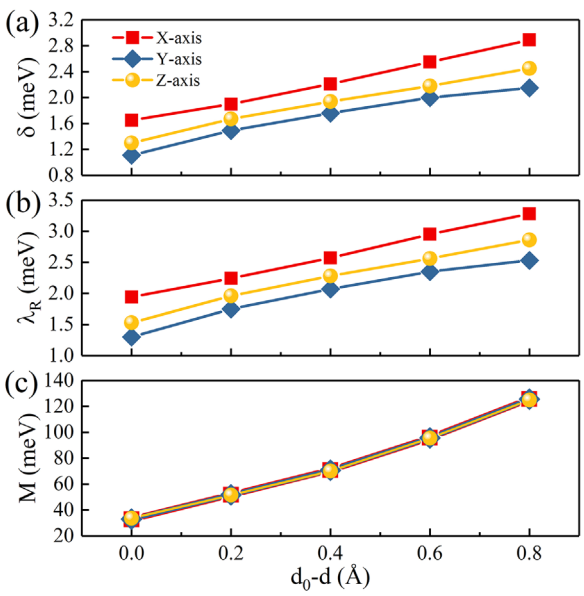

FIG. 6. (a) Band gap δ, (b) the Rashba SOC strength λ_R, and (c) the exchange field M vs the decrease in the BFO/graphene distance with three different magnetizations. The red square, blue diamond, and yellow sphere symbols indicate the data calculated by shifting graphene toward the Fe atoms with magnetizations aligned in the X-axis, Y-axis, and Z-axis, respectively.

In summary, we studied the electronic structures in a BFO/graphene superlattice, where the graphene monolayer is embedded between the slabs of antiferromagnetic $BiFeO_3$ with (111) planes. By calculating the charge density difference and the magnetic properties, we discovered the electron charge transfer from the graphene to Fe atomic layer, leading to the magnetization of graphene with Fermi energy lies at the intersecting points between spin-majority and minority bands. The magnetization direction of the superlattice system is more likely to align in-plane, and the presence of SOC and sublattice potential open band gaps of about 1.65, 1.11, and 1.30 meV with the Fermi level located inside the bandgap and the magnetic moment of Fe atoms in the X-axis, Y-axis, and Z-axis, respectively. It is noteworthy that further consideration of the van der Waals interaction may reduce the distance between graphene and the Fe atomic layer from d_0 = 2.94 Å to d_0 = 2.60 Å in the absence of stress, which helps one to obtain the larger bandgap δ. Our work strongly validates the presence of electronic gaps in the BFO/graphene superlattice, which might be seen in similar superlattices of honeycomb lattices with many other antiferromagnetic perovskites.

See the supplementary material for the total energy with three different magnetization directions and the magnetic anisotropy energy (MAE) within the *xoy* plane of the BFO/graphene superlattice.

This work was supported by the National Natural Science Foundation of China (Grants Nos. 12074241, 52130204, 11929401, and 51861145315), the Shanghai Municipal Science and Technology Commission Program (Grant Nos. 19010500500 and 20501130600), the Independent Research and Development Project of State Key Laboratory of Advanced Special Steel, Shanghai Key Laboratory of Advanced Ferrometallurgy, Shanghai University (No. SKLASS 2020-Z07), the Science and Technology Commission of Shanghai Municipality (No. 19DZ2270200), and the Key Research Project of Zhejiang Laboratory (No. 2021PE0AC02).

AUTHOR DECLARATIONS

Conflict of Interest

The authors declare that they have no competing interests.

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

REFERENCES

[1]E. H. Hall, Am. J. Math. **2**, 287–292 (1879).
[2]Y. Zhang, Y.-W. Tan, H. L. Stormer, and P. Kim, Nature **438**, 201 (2005).
[3]F. D. M. Haldane, Phys. Rev. Lett. **61**, 2015–2018 (1988).
[4]M. Onoda and N. Nagaosa, Phys. Rev. Lett. **90**, 206601 (2003).
[5]C.-Z. Chang, J. Zhang, X. Feng, J. Shen, Z. Zhang, M. Guo, K. Li, Y. Ou, P. Wei, L.-L. Wang, Z.-Q. Ji, Y. Feng, S. Ji, X. Chen, J. Jia, X. Dai, Z. Fang, S.-C. Zhang, K. He, Y. Wang, L. Lu, X.-C. Ma, and Q.-K. Xue, Science **340**, 167–170 (2013).
[6]K. S. Novoselov, A. K. Geim, S. V. Morozov, D. Jiang, M. I. Katsnelson, I. V. Grigorieva, S. V. Dubonos, and A. A. Firsov, Nature **438**, 197 (2005).
[7]A. H. Castro Neto, F. Guinea, N. M. R. Peres, K. S. Novoselov, and A. K. Geim, Rev. Mod. Phys. **81**, 109–162 (2009).
[8]S. D. Sarma, S. Adam, E. H. Hwang, and E. Rossi, Rev. Mod. Phys. **83**, 407–470 (2011).

Published under an exclusive license by AIP Publishing

[9]Y. Li, D. West, H. Huang, J. Li, S. B. Zhang, and W. Duan, Phys. Rev. B **92**, 201403(R) (2015).
[10]X. Deng, S. Qi, Y. Han, K. Zhang, X. Xu, and Z. Qiao, Phys. Rev. B **95**, 121410(R) (2017).
[11]Z. Qiao, W. Ren, H. Chen, L. Bellaiche, Z. Zhang, A. H. MacDonald, and Q. Niu, Phys. Rev. Lett. **112**, 116404 (2014).
[12]J. Zhang, B. Zhao, Y. Yao, and Z. Yang, Phys. Rev. B **92**, 165418 (2015).
[13]M. W. Daniels, W. Yu, R. Cheng, J. Xiao, and D. Xiao, Phys. Rev. B **99**, 224433 (2019).
[14]T. Eelbo, M. Waśniowska, P. Thakur, M. Gyamfi, B. Sachs, T. O. Wehling, S. Forti, U. Starke, C. Tieg, A. I. Lichtenstein, and R. Wiesendanger, Phys. Rev. Lett. **110**, 136804 (2013).
[15]H. Chen, Q. Niu, Z. Zhang, and A. H. MacDonald, Phys. Rev. B **87**, 144410 (2013).
[16]Y. Ren, Z. Qiao, and Q. Niu, Rep. Prog. Phys. **79**, 066501 (2016).
[17]Y. Ren, X. Deng, Z. Qiao, C. Li, J. Jung, C. Zeng, Z. Zhang, and Q. Niu, Phys. Rev. B **91**, 245415 (2015).
[18]S. Baidya, U. V. Waghmare, A. Paramekanti, and T. Saha-Dasgupta, Phys. Rev. B **94**, 155405 (2016).
[19]S. Qi, Z. Qiao, X. Deng, E. D. Cubuk, H. Chen, W. Zhu, E. Kaxiras, S. B. Zhang, X. Xu, and Z. Zhang, Phys. Rev. Lett. **117**, 056804 (2016).
[20]Z. Qiao, Y. Han, L. Zhang, K. Wang, X. Deng, H. Jiang, S. A. Yang, J. Wang, and Q. Niu, Phys. Rev. Lett. **117**, 056802 (2016).
[21]Z. Lin, W. Qin, J. Zeng, W. Chen, P. Cui, J.-H. Cho, Z. Qiao, and Z. Zhang, Nano Lett. **17**, 4013–4018 (2017).
[22]S. Hu, L. Chen, Y. Wu, L. Yu, X. Zhao, S. Cao, J. Zhang, and W. Ren, Chin. Sci. Bull. **59**, 5170–5179 (2014).
[23]D. Xiao, W. Zhu, Y. Ran, N. Nagaosa, and S. Okamoto, Nat. Commun. **2**, 596 (2011).
[24]Y. Weng, X. Huang, Y. Yao, and S. Dong, Phys. Rev. B **92**, 195114 (2015).
[25]Y. Weng, L. Lin, E. Dagotto, and S. Dong, Phys. Rev. Lett. **117**, 037601 (2016).
[26]W. Ren and L. Bellaiche, Phys. Rev. B **82**, 113403 (2010).
[27]W. Ren and L. Bellaiche, Phys. Rev. Lett. **107**, 127202 (2011).
[28]W. Ren, Adv. Manuf. **1**, 166–175 (2013).
[29]W. Ren, Y. Yang, O. Diéguez, J. Íñiguez, N. Choudhury, and L. Bellaiche, Phys. Rev. Lett. **110**, 187601 (2013).
[30]D. Wang, R. Kou, D. Choi, Z. Yang, Z. Nie, J. Li, L. V. Saraf, D. Hu, J. Zhang, G. L. Graff, J. Liu, M. A. Pope, and I. A. Aksay, ACS Nano **4**, 1587–1595 (2010).
[31]A. K. Geim and I. V. Grigorieva, Nature **499**, 419 (2013).
[32]G. Catalan and J. F. Scott, Adv. Mater. **21**, 2463–2485 (2009).
[33]J. F. Ihlefeld, N. J. Podraza, Z. K. Liu, R. C. Rai, X. Xu, T. Heeg, Y. B. Chen, J. Li, R. W. Collins, J. L. Musfeldt, X. Q. Pan, J. Schubert, R. Ramesh, and D. G. Schlom, Appl. Phys. Lett. **92**, 142908 (2008).
[34]J. Wang, J. B. Neaton, H. Zheng, V. Nagarajan, S. B. Ogale, B. Liu, D. Viehland, V. Vaithyanathan, D. G. Schlom, U. V. Waghmare, N. A. Spaldin, K. M. Rabe, M. Wuttig, and R. Ramesh, Science **299**, 1719–1722 (2003).
[35]F. Kubel and H. Schmid, Acta Crystallogr., Sect. B **46**, 698–702 (1990).
[36]R. K. Katiyar, P. Misra, F. Mendoza, G. Morell, and R. S. Katiyar, Appl. Phys. Lett. **105**, 142902 (2014).
[37]Y.-F. Wu, H.-D. Song, L. Zhang, X. Yang, Z. Ren, D. Liu, H.-C. Wu, J. Wu, J.-G. Li, Z. Jia, B. Yan, X. Wu, C.-G. Duan, G. Han, Z.-M. Liao, and D. Yu, Phys. Rev. B **95**, 195426 (2017).
[38]P. Hohenberg and W. Kohn, Phys. Rev. **136**, B864–B871 (1964).
[39]G. Kresse and J. Furthmüller, Phys. Rev. B **54**, 11169–11186 (1996).
[40]J. P. Perdew, K. Burke, and M. Ernzerhof, Phys. Rev. Lett. **77**, 3865–3868 (1996).
[41]S. Grimme, J. Comput. Chem. **25**, 1463–1473 (2004).
[42]C. Chen, L. Fang, G. Zhao, X. Liu, J. Wang, L. A. Burton, Y. Zhang, and W. Ren, J. Mater. Chem. C **9**, 9539–9544 (2021).
[43]S. Hu, H. Gao, Y. Qi, Y. Tao, Y. Li, J. R. Reimers, M. Bokdam, C. Franchini, D. Di Sante, A. Stroppa, and W. Ren, J. Phys. Chem. C **121**, 23045–23054 (2017).
[44]Z. Qiao, S. A. Yang, W. Feng, W.-K. Tse, J. Ding, Y. Yao, J. Wang, and Q. Niu, Phys. Rev. B **82**, 161414 (2010).
[45]Z. Qiao, H. Jiang, X. Li, Y. Yao, and Q. Niu, Phys. Rev. B **85**, 115439 (2012).
[46]H. Jiang, Z. Qiao, H. Liu, J. Shi, and Q. Niu, Phys. Rev. Lett. **109**, 116803 (2012).

Published under an exclusive license by AIP Publishing